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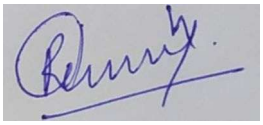
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INSTRUCTIONS:

*1. **Go to File → Download → Microsoft Word (.docx)***
2. Download the IDF in your local storage
3. Explain your invention comprehensively in the form.
4. Save your IDF file with the name of the Title of your invention. eg: "A Smart Electric Kettle"
5. Upload the duly filled IDF below.

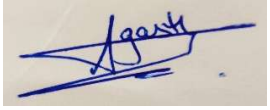
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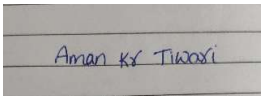
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A) Please enlist the **Key concepts and terms/Keywords/Synonyms used in the Invention**?

S. No	Keywords for invention which has to be search	Synonyms
1	*Wearable Safety Device*	*Smart wearable, safety band*
2	*Panic Detection*	*Fear detection, distress recognition*
3	*Biometric Monitoring*	*Physiological sensing*
4	*SOS Alert System*	*Emergency alert system*
5	*Automatic Trigger*	*Autonomous activation*
6	GPS Tracking	Location tracking
7	Multi-sensor Analysis	Multi-parameter detection

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1. TITLE OF INVENTION- 15 words max

SURAKSHA+ - Intelligent Wearable Panic Detection System
" When panic disables action , technology acts "

2. BACKGROUND OF THE INVENTION- prior art search, tell what was the problem in past and how your invention overcome it. Compare your invention with closet prior art.

Women safety – A Critical Challenge

Women safety has become a critical concern due to increasing cases of harassment and assault. Existing solutions such as mobile safety applications, emergency helpline numbers, smartwatches, and self-defense tools rely heavily on **manual activation by the victim**. However, during panic situations, the human brain undergoes a *fight-or-flight response*, significantly impairing decision-making ability. Victims are often unable to:

- Dial emergency numbers
- Access mobile devices
- Take conscious action

Limitations of Prior Art:

- Safety apps require manual triggering
- Emergency numbers require time-consuming dialing
- Smart devices need conscious user interaction
- Self-defense tools require physical response

Problem:
All existing systems fail when the victim is unable to act.
Solution Provided by the Invention:
The proposed invention overcomes this limitation by introducing an **automatic panic detection system** that works without requiring user input.

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3. OBJECTIVE OF THE INVENTION

- To develop an **automated emergency response system** that functions without manual triggering
- To detect panic situations using **biometric signals**
- To provide **real-time SOS alerts with GPS location**
- To ensure **rapid response during emergencies**
- To reduce dependency on human decision-making under stress
- To minimize false alarms using **multi-factor verification**

4.NOVEL ASPECTS OF THE INVENTION (Point out the new parts used in the invention which make it different from the other existing inventions or prior arts).

- **Automatic Panic Detection:** No manual trigger required
- **Multi-sensor Integration:** Combines heart rate, muscle tension, and motion detection
- **Multi-trigger Verification Logic:** Reduces false alarms (<2%)
- **5-Second Cancel Window:** Prevents false emergency alerts
- **Mandatory Emergency Contact Rule:** Police/helpline cannot be removed
- **Wearable + Mobile + Cloud Ecosystem:** Complete integrated solution
- **Real-time Data Processing via Bluetooth + Mobile App**
  -eSIM-enabled autonomous communication eliminating dependency on smartphones

5. DETAILED DESCRIPTION OF THE INVENTION - *Be sure to use as much space as needed, and be as detailed as you can in this section, as anything that is not covered here will not be protected under the patent we may obtain.*

******Note: please explain your invention step wise in sequential order and how it works and duly explain all the parts of the device in sequential order with its functionality.***

*1. In case of **software embedded with the device**; - explain step wise how the software works with the device and its functionality, with flow chart and diagram. ****
*2. **In case the invention is related to product, device and apparatus, detail of parts of the device, apparatus, product in bullet points***

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3. In case the invention is related to process, composition, explain the steps of process, composition in bullet points

5.1 OVERALL SYSTEM DESCRIPTION

The present invention, titled **"SURAKSHA+ – Intelligent Wearable Panic Detection System"**, relates to a **smart safety system** comprising:

- A **wearable electronic device**
- A **mobile application**
- A **cloud-based backend server**

The system is designed to **automatically detect panic or distress conditions** using real-time biometric and motion data and to **trigger emergency alerts without requiring manual intervention**.

eSIM-Based Communication Module

The proposed system incorporates an embedded SIM (eSIM) module within the wearable device to enable direct and independent cellular communication without relying on a paired smartphone.

Unlike conventional wearable safety systems that depend on Bluetooth connectivity and mobile devices, the integration of eSIM technology allows the device to function autonomously in emergency situations.

In the event of panic detection, the wearable device can directly transmit SOS alerts, including real-time GPS location, to emergency contacts and authorities via cellular networks (4G/5G), ensuring uninterrupted communication even when the user's mobile device is unavailable, switched off, or out of reach.

This significantly enhances the reliability and responsiveness of the system, particularly in critical scenarios where immediate action is required and manual intervention is not possible.

5.2 COMPONENTS OF THE INVENTION (DEVICE / APPARATUS DETAILS)

Hardware Description
A. Wearable Device (Hardware Unit – Detailed Description)

The wearable device is a compact, body-worn safety system designed for continuous physiological monitoring, motion analysis, and autonomous emergency response. The

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hardware architecture is optimized for real-time processing, low power consumption, and multi-mode communication.

The device comprises the following components:

• Microcontroller Unit (ESP32):

The system is built around the ESP32 microcontroller, which serves as the central processing and control unit.
It provides integrated Bluetooth Low Energy (BLE) and Wi-Fi capabilities, enabling seamless communication with the mobile application and cloud services.

The ESP32 is responsible for:

Collecting and processing sensor data
Managing communication modules
Executing panic detection logic at the device level (edge processing)
Controlling alert mechanisms and system operations
• Pulse Rate Sensor (MAX30101 / MAX30102):

The MAX30101/MAX30102 optical sensor is used to continuously monitor the user's pulse rate.
It detects variations in blood flow and identifies sudden spikes associated with stress or panic conditions.

• Motion Sensor (MPU6050):

The MPU6050, which integrates a 3-axis accelerometer and gyroscope, is used to detect motion patterns, sudden jerks, and abnormal movements.
The system utilizes motion data to infer physical distress and unusual activity.

• Bluetooth Low Energy (BLE) Communication:

Enabled via ESP32, BLE is used for low-power, real-time communication between the wearable device and the mobile application.

• eSIM Module (Cellular Communication Unit):

An eSIM module is integrated to provide direct cellular connectivity over 4G/5G networks.
This enables the device to operate independently and transmit emergency alerts without relying on a smartphone.

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• GPS Module:

A dedicated GPS module is used to obtain real-time location data, which is included in emergency alerts and live tracking features.

• Vibration Motor (Haptic Feedback Unit):

Provides tactile feedback during panic detection and the 5-second cancellation window.

• Power Supply Unit (Battery + Power Management System):

The device is powered by a rechargeable battery with an integrated power management system for efficient energy utilization.
It also supports low-battery detection, enabling preventive alert notifications.

B. Mobile Application (Software Unit)

The mobile application acts as the **central processing and control system**, containing:

- **Data Processing Engine**
- **Panic Detection Algorithm (Multi-trigger logic)**
- **User Interface (UI)**
- **Contact Management Module**
- **GPS Location Module**
- **Alert Transmission System**

C. Cloud Server (Backend System)

- **Database Storage** for logs and history
- **API Gateway** for communication
- **Security Layer** for encryption and authentication
- **Alert Routing Mechanism**

5.3 SEQUENTIAL WORKING OF THE INVENTION (STEP-BY-STEP)

Step 1: Initialization & Pairing

- The wearable device is paired with the mobile application via Bluetooth.
- User registers and configures emergency contacts.

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- A mandatory emergency contact (Police/Helpline) is pre-configured and cannot be removed.

Step 2: Continuous Data Acquisition

- Sensors continuously collect:
  - Heart rate data
  - Muscle tension signals
  - Motion/acceleration data
- Data is sampled at predefined intervals and transmitted to the mobile app.

Step 3: Data Transmission

- Sensor data is transmitted via **Bluetooth Low Energy (BLE)** in real-time.
- Data packets include timestamped biometric readings.

Step 4: Data Preprocessing

- Noise filtering is applied to remove irregular spikes.
- Data normalization is performed.
- Baseline values for each user are considered.

Step 5: Panic Detection Algorithm (Core Logic)

The system uses a **multi-trigger verification mechanism**, where panic is detected only when multiple conditions are satisfied:

Trigger Conditions:

- Heart rate exceeds threshold (e.g., sudden spike above baseline)
- Muscle tension increases significantly
- Sudden jerk or abnormal movement detected

Decision Logic:

- Single trigger → Ignored (to avoid false alarm)
- Multiple triggers → Panic condition confirmed

Step 6: Panic Confirmation & User Verification

- When panic is detected:
  - Wearable device activates vibration motor
  - A 5-second cancel window is initiated
- User can:

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- Cancel via wearable (gesture/button, if implemented)
- Cancel via mobile application

Step 7: Automatic SOS Activation
If no cancellation occurs within 5 seconds:

- SOS alert is automatically triggered
- No user interaction required

Step 8: The system transmits emergency alerts through multiple communication channels:
Primary Mode: eSIM-based direct cellular communication (4G/5G)
Secondary Mode: Mobile application via internet (if connected)

This dual-layer communication ensures high reliability and continuous connectivity during emergency situations.

Step 9: Live Location Tracking

- GPS tracking is activated
- Location updates are continuously shared until manually stopped

Step 10: Data Logging & Storage

- Incident details stored in:
  - Mobile application logs
  - Cloud database

5.4 SOFTWARE FUNCTIONALITY (EMBEDDED + APPLICATION LEVEL)
Software Workflow:

1. Receive sensor data from wearable
2. Preprocess and clean data
3. Apply panic detection algorithm
4. Evaluate multi-trigger conditions
5. Trigger verification stage (5-sec window)
6. Execute SOS protocol
7. Send alerts via internet (4G/5G/Wi-Fi)
8. Track and transmit GPS location

5.6 PROCESS DESCRIPTION (PROCESS CLAIM SUPPORT)
The invention also defines a process for automatic panic detection, comprising:

- Collecting biometric data using wearable sensors
- Transmitting data to a mobile application
- Processing data using a multi-trigger algorithm

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- Detecting panic condition based on combined signals
- Providing a user verification window
- Automatically sending SOS alerts upon non-cancellation
- Sharing real-time location with emergency contacts

5.7 SAFETY AND RELIABILITY FEATURES

- Multi-factor authentication of panic signals
- False alarm minimization (<2%)
- Manual override option
- Mandatory emergency contact inclusion
- Secure data transmission

5.8 SCOPE OF THE INVENTION

The invention is not limited to:

- Women safety only
- Wrist wearable only

It can be extended to:

- Elderly care systems
- Child safety devices
- Medical emergency detection systems

Future Scope of the Invention

The proposed invention is designed with a modular and scalable architecture, allowing seamless integration of advanced technologies in future developments. The following enhancements define the extended scope of the system:

• Satellite Communication Capability:

The system can be extended to support satellite-based communication, enabling functionality in remote or no-network areas where conventional cellular connectivity is unavailable. This ensures uninterrupted emergency communication regardless of geographical limitations.

• LoRa-Based Long-Range Communication:

Future implementations may incorporate long-range, low-power communication technologies to enable device-to-device or device-to-gateway communication over extended distances. This is particularly useful in rural areas, disaster zones, and low-connectivity environments.

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• Hybrid Communication Architecture:

The system can evolve into a multi-layer communication framework combining short-range (BLE), cellular (eSIM), long-range (LoRa-like), and satellite communication.

This architecture enables automatic switching between available communication modes, ensuring maximum reliability and continuous connectivity during emergencies.

• Advanced Intelligent Assistance:

The system can be further enhanced with advanced intelligent modules capable of providing contextual assistance, predictive alerts, and improved interaction with users based on system conditions and usage patterns.

• Miniaturization and Form Factor Expansion:

The wearable device can be further miniaturized and integrated into multiple form factors such as rings, pendants, smart clothing, or other accessories, improving user comfort and adoption.

• Integration with Public Safety Infrastructure:

The system can be integrated with public safety networks, enabling direct communication with emergency services, law enforcement agencies, and nearby response units for faster and more efficient intervention.

• Crowd-Assisted Emergency Response:

Future scope includes enabling anonymized alert sharing with nearby users or devices within a secure network, allowing faster local assistance before official responders arrive.

• Extension to Medical and Health Monitoring:

The system can be expanded to detect medical emergencies such as sudden health deterioration, falls, or abnormal physiological conditions, making it applicable for elderly care and health monitoring.

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6. DRAWINGS OR SUPPORT MATERIAL

Step No.	Scenario: Device Connected via BLE (Smartphone Available)	Scenario: Device NOT Connected (eSIM Active Mode)
1	Wearable device detects panic using sensors	Wearable device detects panic using sensors
2	Sensor data transmitted to mobile app via Bluetooth Low Energy (BLE)	Panic detected internally by onboard MCU (edge processing)
3	Mobile application processes data using panic detection algorithm	On-device logic confirms panic using pre-configured thresholds
4	5-second cancel window triggered via mobile app + wearable vibration	5-second cancel window triggered via wearable vibration
5	If not cancelled, mobile app initiates SOS protocol	If not cancelled, wearable device initiates SOS protocol
6	Mobile app fetches GPS location from smartphone	Wearable device uses built-in GPS module
7	SOS alert sent via internet (4G/5G/Wi-Fi) through smartphone	SOS alert sent directly via eSIM cellular network (4G/5G)
8	Alerts delivered to emergency contacts and authorities	Alerts delivered to emergency contacts and authorities
9	Continuous live tracking via mobile app	Continuous live tracking via direct eSIM communication
10	Data logged in mobile app + cloud server	Data logged directly to cloud server via cellular network

The system ensures uninterrupted emergency communication by dynamically switching between BLE-assisted smartphone mode and eSIM-based autonomous mode, thereby maintaining functionality even in the absence of a mobile device.

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7. ABSTRACT (150 words)

Suraksha+ is an intelligent wearable panic detection system designed to enhance personal safety by eliminating the need for manual intervention during emergencies. The system integrates a wearable device with a mobile application and cloud infrastructure to detect panic situations using biometric signals such as heart rate, muscle tension, and sudden motion. Unlike traditional safety solutions, the proposed invention operates automatically using a multi-trigger verification algorithm, ensuring high accuracy and minimal false alarms. Upon detecting a panic event, the system provides a 5-second cancellation window, after which an SOS alert is automatically sent to predefined contacts and emergency services along with real-time GPS location. The invention ensures rapid response, continuous monitoring, and enhanced reliability. This system is particularly beneficial in situations where the victim is unable to act, thereby bridging the gap between panic and protection through intelligent automation.

8. BRIEFLY EXPLAIN THE STEPS, FLOW CHART, FUNCTIONALITY OF INVENTION RELATED TO SOFTWARE, EMBEDDING WITH MACHINE, DEVICE, APPRATUS. ***Note: Explain the functionality of software embedded with the device***

Overview

The invention integrates a wearable device with a mobile application to automatically detect panic using biometric data and trigger emergency alerts without user intervention.

Working Steps

Wearable device pairs with mobile application via Bluetooth.
Sensors collect heart rate, muscle tension, and motion data.
Data is transmitted to the mobile application in real time.
Software processes data using a multi-trigger panic detection algorithm.
If panic is detected, a 5-second vibration alert is given.
If not cancelled, an automatic SOS is sent.
Alerts with GPS location are sent to emergency contacts and authorities.

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